Mobile Internet Speed Prediction for Hanoi

COS40007 Artificial Intelligence Engineering

Group Task 1: System Blueprint & Project Planning

Item	Details
Unit	COS40007 Artificial Intelligence Engineering
Institution	Swinburne University of Technology
Task	Group Task 1 (Week 4): System Blueprint & Project Planning
Project	Mobile Internet Speed Prediction for Hanoi
Studio Session No.	[Studio session number]
Group No.	[Given by tutor]
Team members	Nguyễn Nam Khánh ([Student ID]) – Team Leader Hoàng Minh Đức ([Student ID]) Phạm Quang Minh ([Student ID])
Tutor	[Tutor name]
Submission date	[DD/MM/2026]

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging. We are honoured to recognise our connection to Wurundjeri Country, history, culture and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands. We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors, and the Traditional Owners of the lands on which each of us studies and works.

1. Group Agreement and Project Plan

- Studio Session No.: [Studio session number]
- Group No. (given by tutor only): [Given by tutor]
- Name and Students' Numbers of Group Members:

  Nguyễn Nam Khánh ([Student ID]) – Team Leader, Scrum Master; Data Processing, Tester, DevOps

  Hoàng Minh Đức ([Student ID]) – Model Training and Testing, MLOps

  Phạm Quang Minh ([Student ID]) – Frontend, Backend, DevOps

Group style and roles

Who does what, and how will you work together? Who will be responsible for the submitted deliverables? Who will ask questions on behalf of their group? Who is the Scrum Master for each Sprint or all sprints?

Our project builds a web map of Hanoi where a user clicks any point and receives an estimated mobile internet speed (4G and 5G together). The estimate comes from a machine learning model trained on Ookla Open Data speed tests and the distance from each point to 4G/5G cell towers in OpenCelliD.

Working style. We work as a small Scrum team with two-week sprints. Each sprint starts with sprint planning and ends with a sprint review and retrospective, all held in our Tuesday Google Meet meeting. Each member owns one part of the pipeline from start to finish (data, model, application) and DevOps duties are shared, so every member writes code, tests their own work and contributes to the report.

Member	Role	Responsibilities	Deliverables owned
Nguyễn Nam Khánh	Team Leader, Scrum Master Data Processing, Tester, DevOps	Download and clean Ookla tiles and OpenCelliD towers for Hanoi; exploratory data analysis Feature engineering (distance to towers, tower density, 5G share) and the training dataset Test plan, unit tests for data and features, end-to-end testing and bug log GitHub repository, branch protection, pull-request reviews and data versioning Runs meetings, keeps the Trello board up to date, final submission	hanoi_train.csv and data contract Test reports Group agreement and final report
Hoàng Minh Đức	Model Training and Testing, MLOps	3GPP physics module (theoretical speed from tower distances) KNN and Random Forest baselines and the hybrid model (3GPP + XGBoost) with an 80% range Spatial block cross-validation, model comparison and figures Model registry, monitoring, drift detection and the retraining workflow	Saved model bundle Evaluation results MLOps design

Member	Role	Responsibilities	Deliverables owned
Phạm Quang Minh	Frontend, Backend, DevOps	React + Leaflet map with tower nodes and click-to-predict FastAPI backend (/api/predict, /api/towers) and the prediction service that loads the saved model GitHub Actions CI/CD, Docker images and deployment of the demo	Web application CI/CD pipeline API contract

Answers to the questions above

- **Who does what:** each member owns the area in the table above. Work that crosses areas (data columns, model inputs, API format) is agreed in the Tuesday meeting and written in the data and API contracts.
- **Responsible for submitted deliverables:** Nguyễn Nam Khánh combines all parts and submits every deliverable; each member is responsible for the quality of their own section and checks the full report before submission.
- **Asks questions on behalf of the group:** Nguyễn Nam Khánh contacts the tutor; Hoàng Minh Đức is the backup contact when Nguyễn Nam Khánh is not available.
- **Scrum Master:** Nguyễn Nam Khánh is the Scrum Master for all sprints: he runs the meetings, tracks the Trello board and removes blockers.
- **Decision making:** technical decisions inside one area are made by the owner; decisions that affect other members are made together in the Tuesday meeting and recorded in the minutes.

Timeline (Follow Sprint here)

By when do you need to make decisions? When do tasks need to be completed by to ensure you submit on time? (Aim to submit BEFORE the deadline to allow for unforeseen circumstances).

The project runs from Week 4 to Week 12 in five sprints. Each deliverable has an internal deadline two days before the official due date, so there is time for review and for unforeseen problems.

Sprint	Weeks	Goal	Key deliverables	Lead	Decision point
Sprint 0	Week 4	Plan the project	Group agreement and roles System blueprint and MLOps/DevOps design GitHub repository, Trello board, code skeleton	All	Approve scope, roles and tools

Sprint	Weeks	Goal	Key deliverables	Lead	Decision point
Sprint 1	Weeks 5–6	Reliable Hanoi dataset	Ookla tiles for Hanoi (6+ quarters) Hanoi 4G/5G tower table from OpenCelliD EDA notebook hanoi_train.csv and data contract 3GPP physics module with tests (Đức) Map skeleton with mock data (Minh)	Nguyễn Nam Khánh	End of Week 5: enough towers in Hanoi? If not, add OpenStreetMap features and use a coarser grid
Sprint 2	Weeks 7–8	Trained and evaluated models	KNN and Random Forest baselines Hybrid model with 80% range Spatial CV comparison and figures Saved model bundle (v1) API with mock predictor (Minh)	Hoàng Minh Đức	End of Week 8: choose the final model (target: MAE at least 10% lower than the best baseline)
Sprint 3	Weeks 9–10	Working demo	Backend using the saved model Map with tower nodes, click-to-predict and result panel GitHub Actions CI Monitoring and drift scripts (Đức) End-to-end tests (Khánh)	Phạm Quang Minh	End of Week 10: feature freeze, only bug fixes after this
Sprint 4	Weeks 11–12	Release and report	Retraining workflow Deployment and demo video Final report and presentation	Nguyễn Nam Khánh	Final review two days before submission

Key dates

- **Week 5:** decide whether the tower data is good enough; this is the main risk of the project.
- **Week 6:** data freeze: hanoi_train.csv is fixed so model training can start on the same data.
- **Week 8:** model selection: the best model is chosen with spatial cross-validation and saved for the backend.
- **Week 10:** feature freeze: no new features, only testing and bug fixes.
- **Week 12:** all deliverables finished two days before the deadline; the final check is done in the Tuesday meeting.

Communication Plan

Decide and document how your group will communicate and collaborate regularly throughout the project. This could include meeting via Zoom, communicating via a Teams group or Canvas, and using cloud services for shared documents, etc. How often and how will you communicate? What are your expectations of others responding to your communications?

The group uses two communication channels only: a Zalo group chat and a weekly Google Meet meeting.

- **Zalo group chat:** used for all day-to-day communication: progress updates, quick questions, blockers, links to pull requests and meeting minutes. Each member posts a short progress update at least twice a week.
- **Response time:** members reply to Zalo messages within 24 hours. Urgent messages are marked “URGENT” and answered the same day. Blockers are reported within 48 hours instead of waiting for the meeting.
- **Weekly Google Meet meeting:** every Tuesday from 9:00 pm to 10:30 pm. Agenda: (1) each member’s progress on the Trello board, (2) blockers and help needed, (3) decisions, (4) tasks for the next week. In the first meeting of a sprint we do sprint planning; in the last meeting we do the sprint review and retrospective.
- **Minutes:** a note-taker (rotating each week) writes the decisions and action items and posts them in the Zalo group on the same night.
- **Extra meetings:** if a problem cannot wait until Tuesday, a short extra Google Meet call is arranged in the Zalo group.
- **Absence:** a member who cannot attend tells the group at least 24 hours before, posts a written update in Zalo and reads the minutes afterwards.
- **Shared work (tools, not communication channels):** tasks are tracked on the Trello board; code, notebooks and documents are shared in the GitHub repository; large data files are kept in a shared cloud folder because they are too large for Git.

On-track behaviours

What behaviours will result in great group outcomes?

- Attending every Tuesday meeting on time and coming prepared with an update.
- Finishing assigned tasks by the sprint deadline and moving Trello cards as work progresses.
- Raising blockers early on Zalo (within 48 hours) and offering help to other members.
- Reviewing each other’s pull requests constructively and within two days.
- Following the shared data and API contracts, and announcing any change before making it.
- Writing clear commit messages, clean notebooks and short documentation for each part.

Off-track behaviours

What behaviours would result in group outcomes that are not great?

- Missing meetings or deadlines without notice.
- Not replying on Zalo for more than 24 hours during a sprint.

- Pushing directly to the main branch or merging code that has not been tested.
- Changing shared settings (feature list, model inputs, API format) without telling the team.
- Leaving work until the last day, so others cannot review or build on it.
- Doing another member's task without agreement, or criticising people instead of the work.

Resolving tensions

How will we resolve tensions or off-track behaviour as a group?

We resolve problems early and step by step, so small issues do not grow:

- **Step 1 – talk directly:** the members involved discuss the issue privately (Zalo or a short call) within two days.
- **Step 2 – team meeting:** if it is not solved, the issue is raised in the next Tuesday meeting. The team agrees on a fix, for example re-assigning tasks, adjusting a deadline or pairing two members on a task.
- **Step 3 – decision rule:** technical disagreements are settled with evidence where possible (a quick test or a comparison on the data); otherwise by majority vote, with the Team Leader deciding a tie.
- **Step 4 – tutor:** if a member repeatedly does not contribute after Steps 1–3, the Team Leader informs the tutor with evidence from Trello and GitHub.

Team Expectations:

1. Every member completes their Trello tasks by the sprint deadline and reports blockers on Zalo within 48 hours.
2. Every member attends the Tuesday 9:00–10:30 pm Google Meet meeting, or gives 24 hours' notice and a written update.
3. All code reaches the main branch only through a reviewed pull request that passes the automated tests.

2. System Blueprint Diagram: A diagram showing system components, data flow, and ML model integration with an explanation relevant to your applied project.

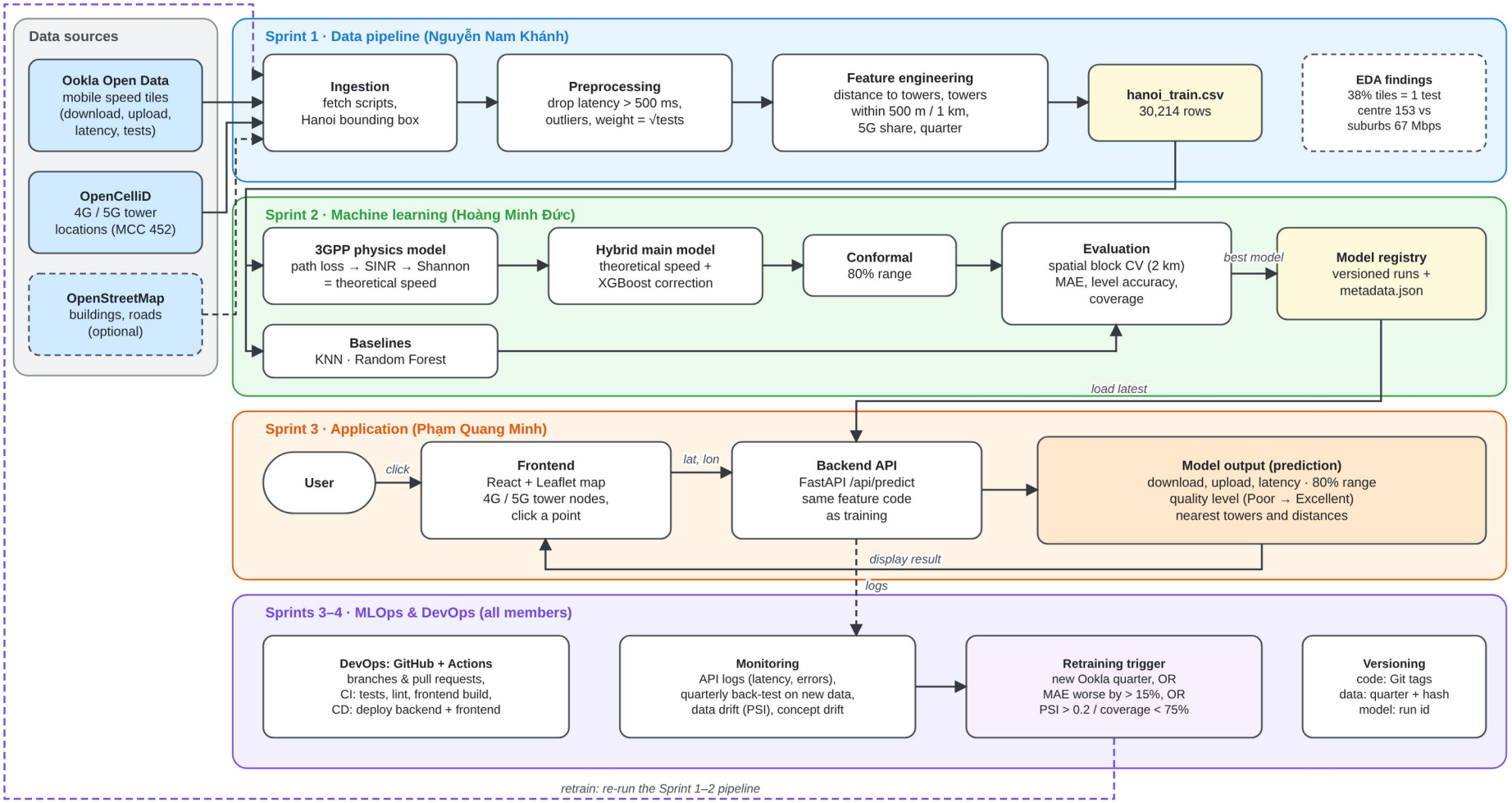


Figure 1. System blueprint. Lanes show the sprint and owner of each stage; solid arrows show data flow, dashed arrows show feedback and optional flows.

System components

- **Data sources:** Ookla Open Data mobile speed-test tiles (about 610 m × 610 m, released every quarter) with download, upload, latency and number of tests; OpenCelliD cell tower locations for Vietnam (MCC 452), filtered to LTE (4G) and NR (5G) towers in Hanoi; OpenStreetMap building data is optional.
- **Ingestion:** scripts download the tiles inside the Hanoi bounding box (latitude 20.53–21.40, longitude 105.28–106.03) and the tower file. The current dataset has 30,214 rows covering 10,459 tiles over six quarters (Q4 2024 to Q1 2026).
- **Preprocessing:** removes invalid rows and extreme values (latency above 500 ms) and adds a sample weight of √tests, because our EDA showed that 38% of tiles contain only one speed test and are noisy.
- **Feature engineering:** for every tile: distance to the nearest tower and the nearest 5G tower, number of towers within 500 m and 1 km, share of 5G towers, the theoretical speed from the physics model and a time index. The same feature code is used for training and by the backend, so both see identical inputs.
- **ML component:** the 3GPP TR 38.901 urban macro model turns tower distances into a theoretical speed (path loss → SINR → Shannon capacity). The hybrid model uses XGBoost to learn the correction between this theoretical speed and the real Ookla speed. KNN (location only) and Random Forest (same features) are baselines. Conformal prediction adds an 80% range to each estimate.
- **Evaluation and model registry:** models are compared with 5-fold spatial block cross-validation (2 km blocks). The best model is saved with its metadata in a versioned model registry, and the backend always loads the latest approved version.
- **Backend (FastAPI):** /api/towers returns the towers to draw on the map; /api/predict receives a latitude and longitude, builds the features, runs the saved model and returns the prediction. Points outside Hanoi are rejected with a clear message.
- **Frontend (React + Leaflet):** shows a map of Hanoi with 4G and 5G towers as coloured nodes. After a click it shows the estimated download, upload and latency, the 80% range, a quality level (Poor below 15 Mbps, Fair 15–100, Good 100–200, Excellent 200 and above) and lines to the three nearest towers.
- **MLOps and DevOps:** request logs and quarterly back-tests feed monitoring; a retraining trigger re-runs the data and training pipeline. GitHub and GitHub Actions provide version control, automated tests and deployment for every stage (Section 3).

Data flow when a user clicks the map

- **1:** the frontend sends the clicked latitude and longitude to /api/predict.
- **2:** the backend checks that the point is inside Hanoi and finds the nearby towers.
- **3:** the shared feature code computes the tower features and the 3GPP theoretical speed.
- **4:** the hybrid model predicts the correction, giving the final estimate and its 80% range.
- **5:** the backend returns the estimate, quality level and nearest towers as JSON; the map shows them and the request is logged for monitoring.

Why this design fits our project. Download speed in Hanoi is spatially clustered (median 153 Mbps within 5 km of Hoan Kiem versus 67 Mbps beyond 20 km), so the model must use location and tower information, and evaluation must use spatial blocks. Speeds also rise over time (median 70.7 Mbps in Q4

2024 to 79.8 Mbps in Q1 2026), so the design includes regular retraining. Ookla tiles are sparse in the suburbs, so the physics part keeps estimates sensible where there is little data.

3. MLOps & DevOps Integration (High-Level Design): How MLOps (retraining, monitoring) and DevOps (CI/CD, version control) will fit into your project.

MLOps

Data and training strategy

- **Unit of data:** one row is one Ookla tile in one quarter (30,214 rows, 10,459 tiles, six quarters).
- **Split:** tiles are grouped into 2 km × 2 km blocks. About 20% of the blocks are held out as the test set and never used for training or tuning. The remaining blocks are used for 5-fold spatial block cross-validation, and 20% of them calibrate the 80% range (about 64% training, 16% calibration/validation, 20% test). We also report results on the newest quarter alone.
- **Why spatial blocks:** neighbouring tiles have very similar speeds; a random split would put near-duplicates in both sets and overstate accuracy.
- **Evaluation targets:** MAE at least 10% lower than the best baseline, at least 60% of quality levels correct, and 75–85% coverage for the 80% range. Metrics use tiles with at least 3 tests; training weights tiles by √tests.

Retraining strategy

Trigger	Rule	Reason
Time-based	Every new Ookla quarter (every 3 months); tower features refreshed monthly	Mobile speeds in Hanoi rise quickly as 5G expands, so the model ages even without errors
Performance-based	The current model is tested on the new quarter first. Retrain if MAE is more than 15% worse than the recorded test MAE, or if 80%-range coverage is outside 75–85%	Detects when predictions become unreliable
Drift-based	Retrain and investigate if PSI > 0.2 for download speed or a key feature	Detects changes in the data before errors become visible

- **Retraining pipeline:** download the new quarter → rebuild features → retrain all models with the same settings → evaluate on the held-out blocks → save as a new version.
- **Champion and challenger:** the retrained model (challenger) replaces the current model (champion) only if it scores better on the same test blocks; otherwise the current model stays in use and the result is logged. This means about four retraining runs per year, plus extra runs from the other triggers.

Monitoring

- **Service monitoring:** the backend logs every request (time, location, prediction, response time, errors). We track response time (target under 1 second), error rate, number of requests and the share of clicks outside Hanoi.
- **Model performance:** users do not report their real speed, so performance is measured with a quarterly back-test: when a new Ookla quarter is released, predictions for those tiles are compared with the measured values (MAE, level accuracy, range coverage) and stored in a metrics history.
- **Data drift:** the new data is compared with the training data using the Population Stability Index (PSI) and the Kolmogorov–Smirnov test for download speed and the main features (tower distance, towers within 1 km, 5G share). PSI above 0.2 is treated as significant drift.
- **Concept drift:** the relationship between features and speed can change, for example when towers are upgraded to 5G. We detect this by tracking the residuals (prediction minus actual) by distance band and quality level, the gap between theoretical and real speed, and changes in XGBoost feature importance.
- **Model versioning:** each trained model is saved in its own folder (artifacts/models/<name>/<run id>) with metadata recording the Git commit, data version, parameters and metrics. A LATEST pointer marks the model used by the backend, so any version can be reproduced or rolled back.

DevOps

- **Version control:** one GitHub repository for code, notebooks and documentation. The main branch is protected; work happens on feature branches (e.g. feature/data-tower-features) and is merged through a pull request reviewed by at least one other member. Commit messages reference the Trello card (e.g. “T-04: add tower features”) and each sprint ends with a release tag (v0.1-sprint1, …).
- **CI pipeline (GitHub Actions):** on every pull request: install dependencies, run the linter, run the unit tests (physics worked example, feature columns, model save/load), check that notebooks have no stored outputs and build the frontend. A pull request cannot be merged if any check fails.
- **CD pipeline:** on merge to main: build Docker images for the backend and frontend and deploy the demo. A scheduled workflow each quarter downloads the new Ookla data, runs the back-test and drift checks, and starts the retraining pipeline if a trigger fires.
- **Data versioning:** data files are kept outside Git in a shared cloud folder and identified by quarter and a SHA-256 hash, which is stored in the model metadata.
- **Ownership:** Hoàng Minh Đức owns retraining, monitoring and the model registry; Phạm Quang Minh owns CI/CD, Docker and deployment; Nguyễn Nam Khánh owns the repository, branch rules, pull-request reviews and tests.

4. Project Management Setup (Agile Tools): A Jira (or Trello or others) board with key user stories and a Confluence/Notion/Word page for project documentation.

Tool: Trello – https://trello.com/b/Cb1sJZPX/cos40007-hanoi-mobile-speed-prediction

- **Lists:** Backlog (all user stories and future tasks) → Sprint To Do (tasks of the current sprint) → In Progress → In Review (pull request open) → Done (merged and tested).
- **Cards:** each user story (US1–US6) is a card with its acceptance criteria and a checklist linking to its tasks. Each task card (T-01 to T-23) has the story points in its title and the user story, owner, priority, sprint and a “done when” condition in its description.
- **Labels:** green = Sprint 1, yellow = Sprint 2, orange = Sprint 3, purple = Sprint 4, red = highest-priority tasks on the critical path (T-02, T-11, T-14), blue = user story.
- **Workflow:** in sprint planning, tasks move from Backlog to Sprint To Do and are assigned to members. A member moves a card to In Progress when starting, to In Review when the pull request is opened, and to Done after the review and tests pass.

Key user stories

ID	User story	Acceptance criteria	Priority	Sprint	Owner
US1	As a mobile user in Hanoi, I want to click any point on a map and see the estimated download speed, so that I can choose where to work or study with good internet.	A click inside Hanoi returns download, upload and latency estimates in under 1 second; a click outside Hanoi shows a clear warning.	High	3	Phạm Quang Minh
US2	As a mobile user, I want each estimate to show a confidence range and a quality level, so that I know how much I can trust it.	Every estimate has an 80% range with 75–85% coverage on test data; the level is shown as Poor, Fair, Good or Excellent.	High	2–3	Hoàng Minh Đức, Phạm Quang Minh
US3	As a mobile user, I want to see the nearest 4G/5G towers and their distances, so that I understand why an area is fast or slow.	4G and 5G towers are shown in different colours; after a click, lines to the three nearest towers show their distances.	Medium	3	Phạm Quang Minh
US4	As a data engineer, I want a clean, reproducible Hanoi dataset with tower features, so that every model trains on the same inputs.	hanoi_train.csv follows the data contract, can be rebuilt from raw data with one notebook, and the feature tests pass.	High	1	Nguyễn Nam Khánh
US5	As an ML engineer, I want to compare the baselines with the hybrid model using spatial cross-validation, so that we can show the improvement honestly.	5-fold spatial block CV for all three models; hybrid MAE at least 10% lower than the best baseline (or the gap is explained); comparison table saved.	High	2	Hoàng Minh Đức
US6	As the maintenance team, I want automated tests, versioned models and a retraining check for each new data quarter, so that the system stays accurate.	CI runs on every pull request; every saved model has metadata; the quarterly back-test and drift check decide whether to retrain.	Medium	3–4	All

Task breakdown

Card	Story	Task	Assignee	Priority	Sprint	Points
T-01	US4	Download Ookla tiles for Hanoi, including newest quarters	Nguyễn Nam Khánh	High	1	2
T-02	US4	Download OpenCelliD (MCC 452), filter Hanoi 4G/5G towers	Nguyễn Nam Khánh	Highest	1	3
T-03	US4	EDA notebook: distributions, map, tests per tile	Nguyễn Nam Khánh	High	1	3

Card	Story	Task	Assignee	Priority	Sprint	Points
T-04	US4	Cleaning module and tower-distance features	Nguyễn Nam Khánh	High	1	5
T-05	US4	Build hanoi_train.csv and data contract; feature tests	Nguyễn Nam Khánh	High	1	3
T-06	US5	3GPP physics module with worked-example tests	Hoàng Minh Đức	High	1	5
T-07	US6	GitHub repo, branch protection, PR template, README	Nguyễn Nam Khánh	High	1	2
T-08	US1	Frontend map skeleton with mock JSON	Phạm Quang Minh	Medium	1	3
T-09	US5	Spatial block CV and metrics	Hoàng Minh Đức	High	2	3
T-10	US5	Train KNN and Random Forest baselines	Hoàng Minh Đức	High	2	3
T-11	US2	Hybrid model (3GPP + XGBoost) and conformal 80% range	Hoàng Minh Đức	Highest	2	5
T-12	US6	Model registry: versioned runs with metadata	Hoàng Minh Đức	Medium	2	2
T-13	US1	FastAPI endpoints with mock predictor	Phạm Quang Minh	High	2	3
T-23	US1	API contract and backend integration tests	Phạm Quang Minh	Medium	2	3
T-14	US1	Predictor that loads the saved model and shared features	Phạm Quang Minh	Highest	3	3
T-15	US3	Tower layer and lines to nearest towers	Phạm Quang Minh	Medium	3	3
T-16	US2	Result panel: estimate, range, quality level	Phạm Quang Minh	High	3	2
T-17	US6	GitHub Actions CI: lint, tests, frontend build	Phạm Quang Minh	Medium	3	3
T-18	US6	Monitoring: request logs, PSI drift, quarterly back-test	Hoàng Minh Đức	Medium	3	3
T-19	US1	End-to-end test plan and bug log	Nguyễn Nam Khánh	High	3	3
T-20	US6	Scheduled retraining with champion/challenger	Hoàng Minh Đức	Medium	4	3
T-21	US1	Docker images, deployment and demo video	Phạm Quang Minh	Medium	4	3
T-22	—	Final report, presentation and submission	Nguyễn Nam Khánh	High	4	5

Workload: Nguyễn Nam Khánh 26 points, Hoàng Minh Đức 24 points and Phạm Quang Minh 23 points, so the work is balanced. Tasks on the critical path (T-02 tower data, T-11 hybrid model, T-14 predictor) have the highest priority because later work depends on them.

[Insert screenshot of the Trello board here]

Project documentation

Location: [Notion / Word page link]. The same content is kept in the GitHub repository (README.md and docs/).

- **Project description:** goal, scope (mobile internet in Hanoi, 4G and 5G together), users and success criteria.
- **System overview:** the blueprint in Figure 1 and how to run the pipeline, backend and frontend.
- **Data contract:** the columns of hanoi_train.csv, their units and meaning, shared between data and model work.
- **API contract:** the request and response JSON of /api/predict and /api/towers, shared between backend and frontend.
- **Design decisions:** data sources, choice of the hybrid model and baselines, spatial cross-validation, the 80% range and the retraining policy, each with its reason.
- **Meeting minutes:** decisions and action items from every Tuesday meeting.

5. Contribution Statement

Each member contributed to Group Task 1 in their own area, and the work was shared equally. All members reviewed the full report before submission.

Member	Role	Contribution to Group Task 1	Share	Signature
Nguyễn Nam Khánh	Team Leader, Data Processing, Tester, DevOps	Wrote the group agreement, roles, sprint plan and communication plan Exploratory data analysis of the Hanoi Ookla data used in the blueprint and design decisions Set up the GitHub repository and the Trello board; wrote the data and testing user stories and tasks Combined, formatted and submitted the report	33.3%	
Hoàng Minh Đức	Model Training and Testing, MLOps	Designed the ML component of the blueprint (3GPP physics, hybrid model, baselines, 80% range) Wrote the MLOps design: data split, retraining triggers, monitoring, data and concept drift, model versioning Wrote the model-related user stories and tasks	33.3%	
Phạm Quang Minh	Frontend, Backend, DevOps	Designed the application part of the blueprint (FastAPI backend, React + Leaflet map) Wrote the DevOps design: version control workflow, CI/CD pipeline, Docker and deployment Wrote the application user stories and tasks	33.3%	

We confirm that the contribution shares above are accurate and agreed by all members.

6. Generative AI Declaration

Our group used generative AI in preparing this submission, as allowed by the unit. We used Claude (Anthropic) for the purposes listed below. All AI output was reviewed, edited and checked by the group against our own data analysis and design decisions, and the group takes full responsibility for the final submission. All numbers from our dataset (row counts, median speeds) were computed by the group from the Ookla data.

Tool	What we used it for	How we checked and changed the output
Claude (Anthropic)	Structuring the report and improving the wording of our own content	Every section was rewritten where needed to match our decisions; facts and numbers were checked by the group

Tool	What we used it for	How we checked and changed the output
Claude (Anthropic)	Suggesting user stories and splitting them into tasks; creating the Trello cards	The team reviewed every story and task, changed owners and story points to match our roles, and agreed them in the Tuesday meeting
Claude (Anthropic)	Producing a first version of the system blueprint diagram from our component list	We checked every component and arrow against our architecture and corrected the layout and labels
Claude (Anthropic)	Explaining MLOps/DevOps concepts (drift tests, CI/CD) and the 3GPP propagation formula	Checked against unit materials and the 3GPP TR 38.901 standard; thresholds were chosen by the group

Proof: the conversation log with the prompts and AI responses is available at [shared conversation link / Appendix A screenshots].